YAKEEN

NEET 2.0 2027

Physics | Lecture - 3

Basic Maths and Calculus

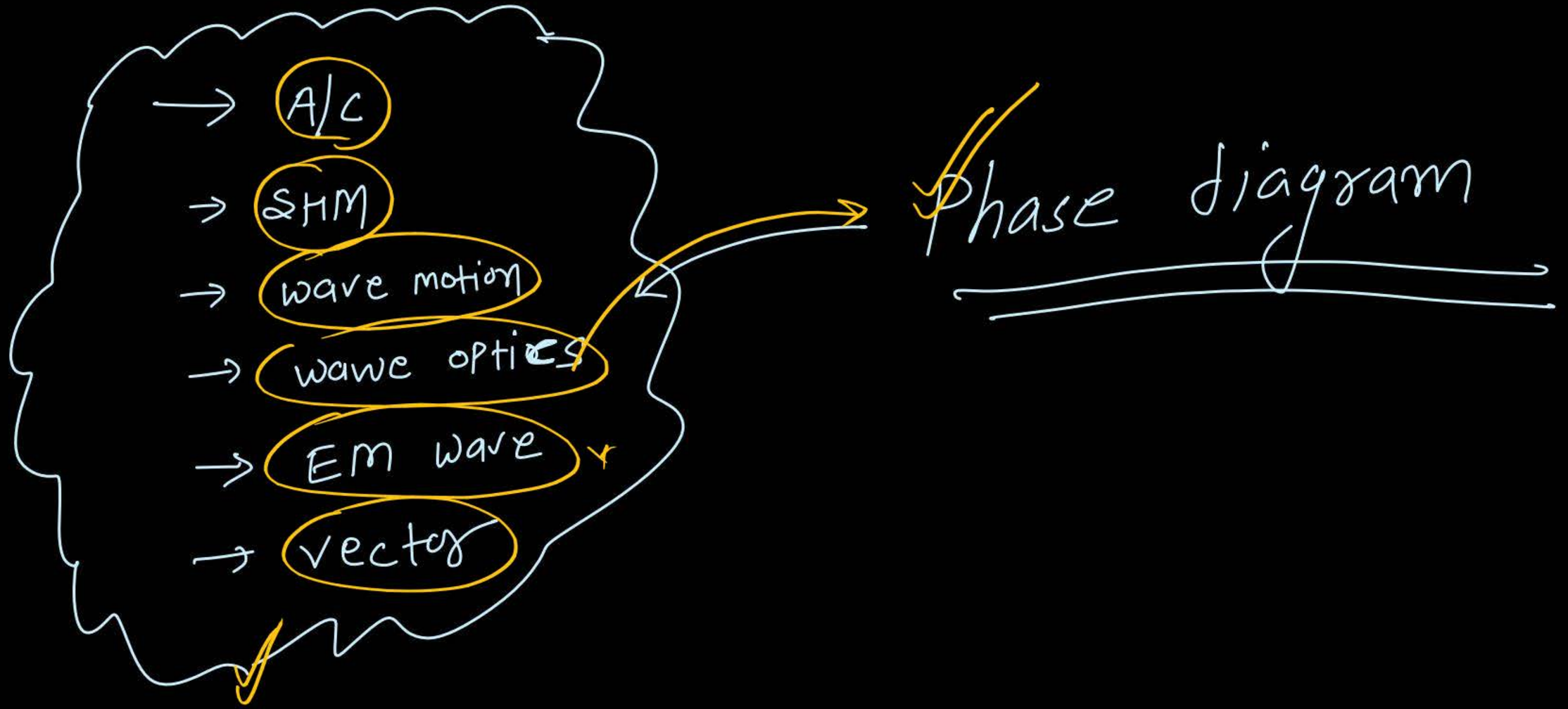
By- Manish Raj (MR Sir)



Todays Goal

ASTC Rule, Phasor diagram.

all H/W we will do today.


A/C
SHM
wave motion
wave optics
EM wave
vector
Phase diagram

$$\frac{3l}{2} = 10 + \frac{l}{2}$$

$$\frac{3l}{2} = 10 + \frac{l}{2}$$

find l

$3l = 10 + l$

$3l - l = 10$

$2l = 10$

$l = 5$ ⟹ correct

$$3 = \frac{2 + \frac{l}{2}}{5}$$

$$3 = \frac{2 + l}{2 \times 5}$$

$$3 = \frac{2 + l}{10}$$

$30 = 2 + l$

$l = 30 - 2 = 28$

$\tan 60° = \frac{10+a}{b}$ $\quad a = \frac{l}{2}, \; b = \frac{l\sqrt{3}}{2}$

(I) $\sqrt{3} = \frac{(10+l)2}{2 \times l\sqrt{3}}$

$2 \times l\sqrt{3} \times \sqrt{3} = (10+l)2$

$2 \times 3l = 20 + 2l$

$6l = 20 + 2l$

$4l = 20$

$l = \frac{20}{4} \; 5$

(II) $\sqrt{3} = \frac{10+l}{\frac{l\sqrt{3}}{2}}$

$\frac{\sqrt{3} \times l\sqrt{3} \times 2}{2} = 10+l$

$\frac{2 \times 3l}{2} = 10+l$

$3l = 10+l$

$3l - l = 10$

$2l = 10$

$l = \frac{10}{2} \; 5$

Hello sir
Me live class nhi kar paya tha part time job ki wajah se.
Mene abhi lecture dekha.
Or mera bhi
5m answer aa rha hai..
Agar meri calculation me koi glti hai to plzz mujhe btaiye taaki me exam me us glti ko na dohrau...
Agar nhi hai to mujhe khushi hogi ke me select hone waale 6.24% me aaya
😊😊

@mrsir_mrstar

"Freedom comes only when we forget our ideals and objectives and principles." ~Jawaharlal Nehru

$\tan 60° = \frac{10+a}{l}$

$\sin 30° = \frac{a}{l}$

$a = \frac{1}{2} l$

$\cos 30° = \frac{b}{l}$

$b = \frac{\sqrt{3}}{2} l$

$\sqrt{3} = \frac{10 + \frac{1}{2} l}{\frac{\sqrt{3}}{2} l}$

$\frac{3}{2} l = 10 + \frac{1}{2} l$

$3l = 10 + l$

$3l - l = 10$

$2l = 10$

$l = 5$

Small Angle approximation

$\cos\underset{\text{small}}{\theta} = 1$

$\cos(2°) = 1$

$\cos(3°) = 1$

$\cos(4°) = 1$

$\sin(\theta) = \theta$ ← must be in radian

$\theta \to$ small

$\tan(\theta) = \theta$

$\theta \to$ small

$\tan(3°) = \sin(3°) = 3°$ *

$= 3 \times \left(\frac{\pi \text{ rad}}{180}\right)$

** COPY

$$\lim_{\theta \to \text{small}} \frac{\sin(\theta)}{\theta} = \frac{\theta}{\theta} = 1$$

Question

copy

Find value:

(i) $\sin 2^\circ = 2^\circ = \frac{2 \times \pi \text{ rad}}{180} = \frac{\pi}{90} \text{ rad}$

(ii) $\tan 3^\circ = \frac{3 \times \pi}{180} = \frac{\pi}{60} \text{ rad}$

(iii) $\cos 3^\circ = 1$

Question

Which of the following option is correct for the value of tan (2°)

1. 2°
2. π/90
3. π/30
4. 2 rad

$$\tan(\theta) = \theta$$

$\tan(2^\circ) = 2^\circ$

H|W

$\sin(105^\circ) = \sin(60^\circ + 45^\circ)$

$= \sin 60 \cos 45 + \sin 45 \cos 60$

$= \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \times \frac{1}{2}$

$= \frac{\sqrt{3}+1}{2\sqrt{2}}$

$\sin(75^\circ) = ?$

$= \sin(45^\circ + 30)$

$= \frac{\sqrt{3}+1}{2\sqrt{2}}$

$$\sin(74^\circ) = \sin(2\times 37^\circ) = 2\sin(37^\circ)\cos(37^\circ)$$

$$= 2\times\frac{3}{5}\times\frac{4}{5} = \frac{24}{25}$$

use:

$$\sin(2\theta) = 2\sin\theta\cdot\cos\theta$$

$$\cos(15^\circ) = \cos(60-45^\circ) = \cos(45-30)$$

$$= \frac{\sqrt{3}+1}{2\sqrt{2}}$$

~~$\sin(105^\circ)$~~

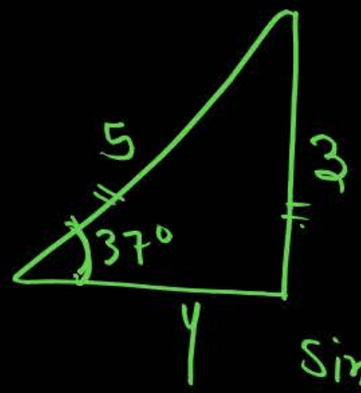
$$\cos(105) = \cos(60+45)$$

$$= \cos 60 \cos 45 - \sin 60 \sin 45$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}}$$

$$= \frac{1-\sqrt{3}}{2\sqrt{2}}$$

$\sin 37^\circ = \frac{3}{5}$



$\sin 37^\circ = \frac{3}{5}$

$\cos 37^\circ = \frac{4}{5}$

Copy

$\sin(22.5^\circ)$

θ

$$\sin\theta = \sqrt{\frac{1-\cos(2\theta)}{2}}$$

$$\sin(22.5^\circ) = \sqrt{\frac{1-\cos(2\times 22.5)}{2}}$$

$$= \sqrt{\frac{1-\cos(45)}{2}} = \sqrt{\frac{1-\frac{1}{\sqrt{2}}}{2}}$$

$$= \sqrt{\frac{\sqrt{2}-1}{2\sqrt{2}}}$$

Kal kiya tha

$$\cos(2\theta) = \cos^2\theta - \sin^2\theta$$

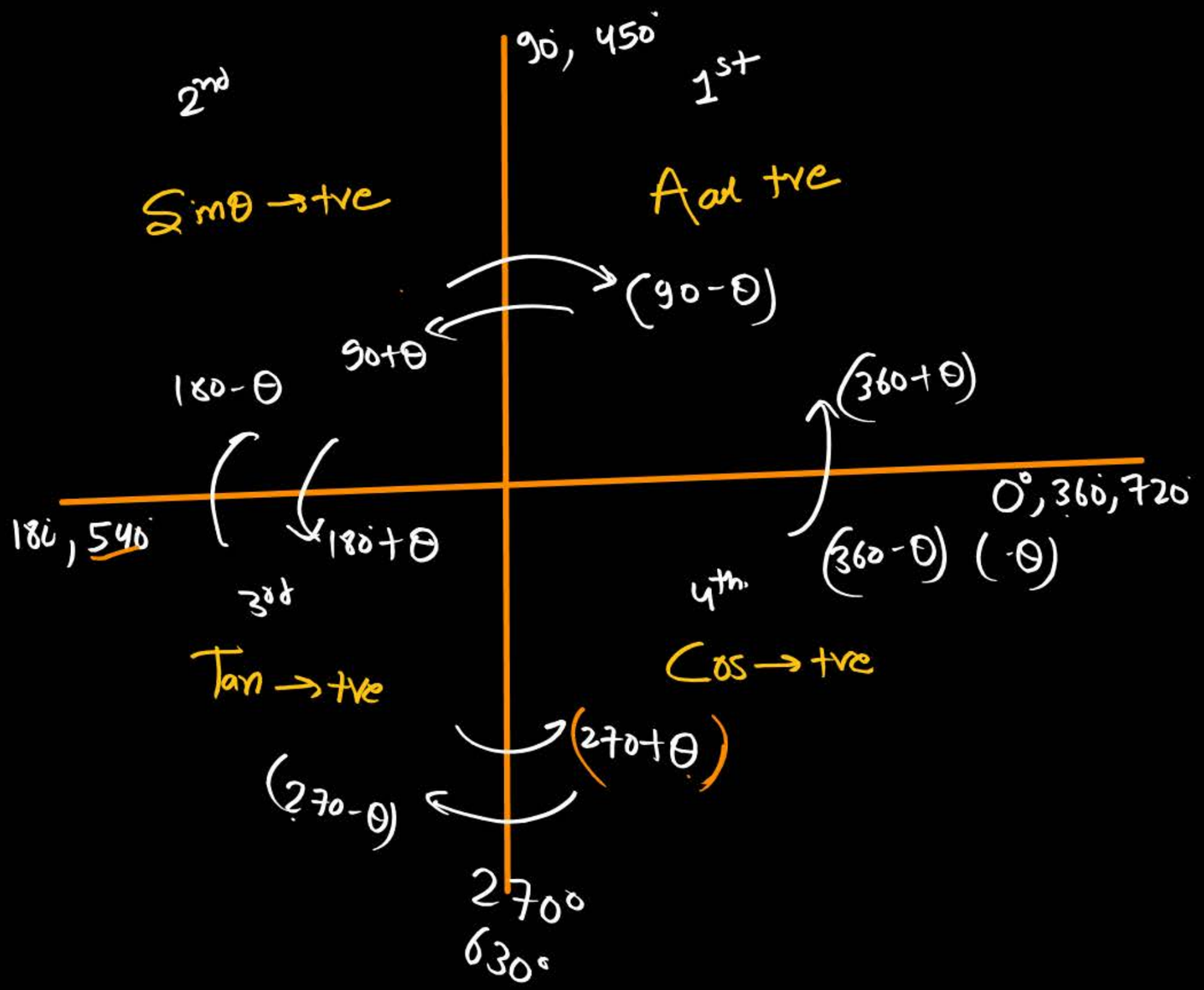
$$\cos(2\theta) = 1 - 2\sin^2\theta$$

$$2\sin^2\theta = 1 - \cos(2\theta)$$

$$\sin\theta = \sqrt{\frac{1-\cos(2\theta)}{2}}$$

ASTC → Rule

given $\theta < 90°$



$$\sin(90+\theta) = +\cos\theta$$

2nd

MR scam Result kya sign Nahi dekhna

$$\sin(90+\theta) = \sin 90 \cos\theta + \sin\theta \cos 90$$

$$= 1 \cos\theta$$

$$= \cos\theta$$

Copy

MR* Box → for ASTC Rule

step-1 decide karo Angle kis quadrant me ja raha hai uske according function ka sign likho

step-2 function $(90 \pm \theta)$ me change hoga. $(270 \pm \theta)$ me same hoga $(180 \pm \theta), (360 \pm \theta)$

step-3 $\sin\theta \rightleftarrows \cos\theta$

$\tan\theta \rightleftarrows \cot\theta$

$\sin(90+\theta) = +\cos\theta$

$\sin(90-\theta) = +\cos\theta$

→ $\cos(90^\circ+\theta) = -\sin\theta$

$\cos(90-\theta) = +\sin\theta$

$\tan(90-\theta) = +\cot\theta$

*$\sin(180+\theta) = -\sin\theta$

$\sin(180-\theta) = +\sin\theta$

→ $\cos(180^\circ+\theta) = -\cos\theta$

$\tan(180-\theta) = -\tan\theta$

$\sin(270^\circ+\theta) = -\cos\theta$

$\sin(270^\circ-\theta) = -\cos\theta$

$\cos(270+\theta) = +\sin\theta$

$\tan(360^\circ+\theta) = +\tan\theta$

→ $\sin(450^\circ+\theta) = +\cos\theta$ (450 = 90)

$\cos(360-\theta) = +\cos\theta$

$\cos(-\theta) = \cos\theta$ ←

→ $\tan(540+\theta) = +\tan\theta$ (540 = 180)

Copy

$\sin(127^\circ) = \sin(90+37) = +\cos 37^\circ = \frac{4}{5}$

$\cos(143^\circ) = \cos(90^\circ+53) = -\sin 53^\circ = -\frac{4}{5}$

$\tan(143^\circ) = \tan(90+53) = -\cot 53^\circ = -\frac{1}{\tan 53^\circ} = -\frac{3}{4}$

$\tan\theta = \frac{1}{\cot\theta}$

* $\sin(300^\circ) = \sin(270^\circ+30^\circ) = -\cos 30^\circ = -\frac{\sqrt{3}}{2}$

$= \sin(360-60) = -\sin 60^\circ = -\sqrt{3}/2$

$\cos(240) = \cos(180+60) = -\cos 60^\circ = -\frac{1}{2}$

$= \cos(270^\circ-30] = -\sin 30^\circ = -\frac{1}{2}$

(copy)

$$\sin(405^\circ) = \sin(360^\circ + 45^\circ) = \sin(45^\circ) = \frac{1}{\sqrt{2}}$$

$$\cos(480) = \cos(450^\circ + 30) = \cos(90^\circ + 30) = -\sin 30^\circ = -\frac{1}{2}$$

$$\cos(780) = \cos(720^\circ + 60) = \cos(60^\circ) = \frac{1}{2}$$

$\sin(A+B)$

Copy

$\Rightarrow \sin(-30^\circ) = -\sin 30^\circ = -\frac{1}{2}$ ✓

$\Rightarrow \cos(-60^\circ) = +\cos 60^\circ = \frac{1}{2}$

$\Rightarrow \sin(-120^\circ) = -\sin(120) = -\frac{\sqrt{3}}{2}$

$\Rightarrow \cos(-120^\circ) = \cos(120) = \cos(90+30) = -\sin 30$
$= -\frac{1}{2}$

$\Rightarrow \sin(-150^\circ) = -\sin(150) = -\frac{1}{2}$

MR* Box

$\cos(-\theta) = \cos\theta$

$\sin(-\theta) = -\sin\theta$

$\tan(-\theta) = -\tan\theta$

Question

H/W copy (show)

Suggest suitable match between function given in the first column and its description given in the second column.

$\sin(360^\circ + 30^\circ) = \sin 30 = +\frac{1}{2}$

Column-I		Column-II	
(A)	$\sin(390^\circ)$	(P)	Positive
(B)	$\sin(-30^\circ) = -\sin 30 = -\frac{1}{2}$	(Q)	Negative
(C)	$\cos 120^\circ = -\frac{1}{2}$	(R)	Zero
(D)	$\tan(-120^\circ) = -\tan(120) = -(-\sqrt{3}) = +\sqrt{3}$	(S)	Modulus is greater than one
		(T)	Modulus is less than one

1. A→PT, B →QT, C→QT, D→PS
2. A→PT, B →QS, C→QT, D→PS
3. A→QT, B →QS, C→PT, D→PS
4. A→QS, B →PT, C→QT, D→PS

$\left|-\frac{1}{2}\right| = \frac{1}{2}$ Modulus → magnitude

MR* BOX

$(\sin\theta)_{max} = +1$ at $\theta = 90^\circ$

$(\cos\theta)_{max} = +1$ at $\theta = 0^\circ$

$(\sin\theta)_{min} = -1$ at $\theta = 270^\circ$

$(\cos\theta)_{min} = -1$ at 180°

(Q) $y = 6\sin\theta$ find maximum and minimum value of 'y'

Soln $y = 6 \times \sin\theta$

$y_{max} = 6(\sin\theta_{max})$

$= +6 \times 1$ at $\theta = 90$

$= +6$

$y_{min} = -6$

$\theta = 270$

Copy

(Q) If $y = 4\sin(3\theta)$ then find y_{max} and θ at which y will be maximum.

$y_{max} = 4 \sin(3\theta)_{max}$

Angle

$y_{max} = 4 \times 1$

$= +4$

Angle $= 90^\circ = 3\theta$

$\sin(3\theta)_{max} = +1$

at $\theta = \frac{90}{3} = 30^\circ$

(Q) If $R = \frac{u^2 \sin(2\theta)}{g}$ find θ at which R will be maximum. When u & g = const

$R_{max} = \left(\frac{u^2}{g}\right)_{const} \sin(2\theta)_{max}$

$\sin(2\theta) = +1$

$2\theta = 90$

$\theta = 45^\circ$

$R_{max} = \frac{u^2}{g} \times 1$

$\theta = 45^\circ$

$y = 3\sin\theta + 4\sin\theta$ find y_{max}

$y = 7\sin\theta$ ✓

$y_{max} = 7$ at $\theta = 90^\circ$

(Q) $y = 3 + 2\sin\theta$ find y_{max} / y_{min}.

$$y_{max} = 3 + 2(\sin\theta)_{max}$$

$$= 3 + 2 \times 1 = 5 \quad \text{at } 90^\circ$$

$$y_{min} = 3 + 2(-1) \quad \text{at } 270^\circ$$

$\sin(270) = -1$

$$y_{mi} = 3 - 2$$

$$= +1$$

(8) $y = 3\sin\theta + 4\cos\theta$

$$y_{max} = \sqrt{3^2 + 4^2}$$

$$= \sqrt{9+16}$$

$$= \sqrt{25}$$

$$= +5$$

$$y_{min} = -5$$

MR* Dox

$$y = a\sin\theta + b\cos\theta$$

$$y_{max} = +\sqrt{a^2+b^2}$$

$$y_{min} = -\sqrt{a^2+b^2}$$

Trigonometric function

	maximum y	minimum y
# $y = 3\sin\theta$	$+3$	-3
# $y = 3\cos\theta + 4\cos\theta$ $= 7\cos\theta$	$+7$	-7
# $y = 3 + 4\sin\theta$	$+7$	$y_{min} = 3 + 4(-1)$ $= 3 - 4$ $= -1$
# $y = \sin\theta + 4\cos\theta$ $y = a\sin\theta + b\cos\theta$	$\sqrt{1^2 + 4^2}$ $= \sqrt{1+16} = \sqrt{17}$	$-\sqrt{17}$
# $y = 3\sin\theta - 4$	$y_{max} = -1$ $y_{max} = 3\times1 - 4$ $= 3 - 4$ $= -1$	$y_{min} = -7$ $y_{min} = 3(-1) - 4$ $= -7$

Copy

(9) If $Y = \dfrac{4}{3\sin\theta + 2\cos\theta}$ then find minimum magnitude of 'y'

$$Y_{min} = \frac{4}{(3\sin\theta + 2\cos\theta)_{max}} = \frac{4}{\sqrt{3^2 + (2)2}}$$

$$= \frac{4}{\sqrt{9+4}} = \frac{4}{\sqrt{13}} \checkmark$$

$$\left\{ \frac{4}{10} = 0.4 \right.$$

$$\frac{4}{100} = 0.04$$

$$\left. \frac{4}{1000} = 0.004 \right\}$$

⊙ Phase of Trigonometric function → Phase is Angle Inside Trigonometric function.

$\sin(\theta)$

↑ Phase

$\sin(\theta+\alpha)$

Phase

$\sin(\omega t+\alpha+90^\circ)$

Phase

Phase difference → Angle difference b/w two same trigonometric function is c/d Phase difference.

Ex- $Y_1 = \sin(\theta)$ (ϕ_1)

$Y_2 = \sin(\theta + 60)$ (ϕ_2)

Phase diff^n $\Delta\phi = \phi_2 - \phi_1$

$= (\theta + 60) - \theta$

$= +60°$ (lead) आगे

$\Delta\phi = \phi_1 - \phi_2$

$= \theta - (\theta + 60)$

$= \theta - \theta - 60$

$= -60°$ (lag)

(Q) find Phase diff'

ϕ_1

$Y_1 = \sin(\omega t + 30^\circ)$ आगे

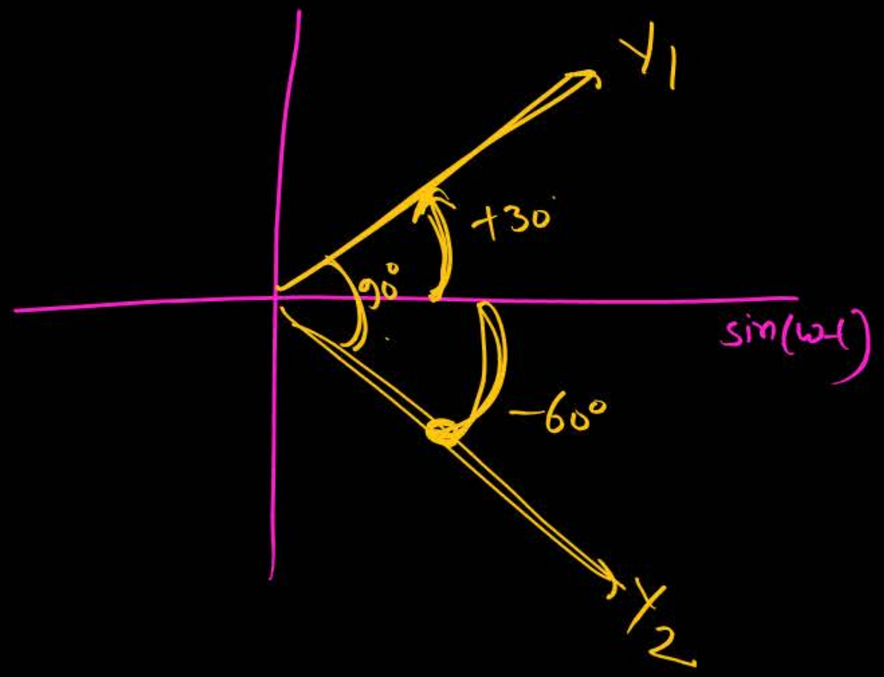
$Y_2 = \sin(\omega t - 60^\circ)$ (पीछे)

$\Delta\phi = \phi_1 - \phi_2$

$= (\omega t + 30) - (\omega t - 60)$

$= \omega t + 30 - \omega t + 60$

$= 90^\circ$



(Q) find phase difference b/w Y_2 and X_1

$$X_1 = A\cos(\omega t - \pi/2)$$

$$Y_2 = A\cos(\omega t + \pi/3)$$

$\pi/3 = 60°$

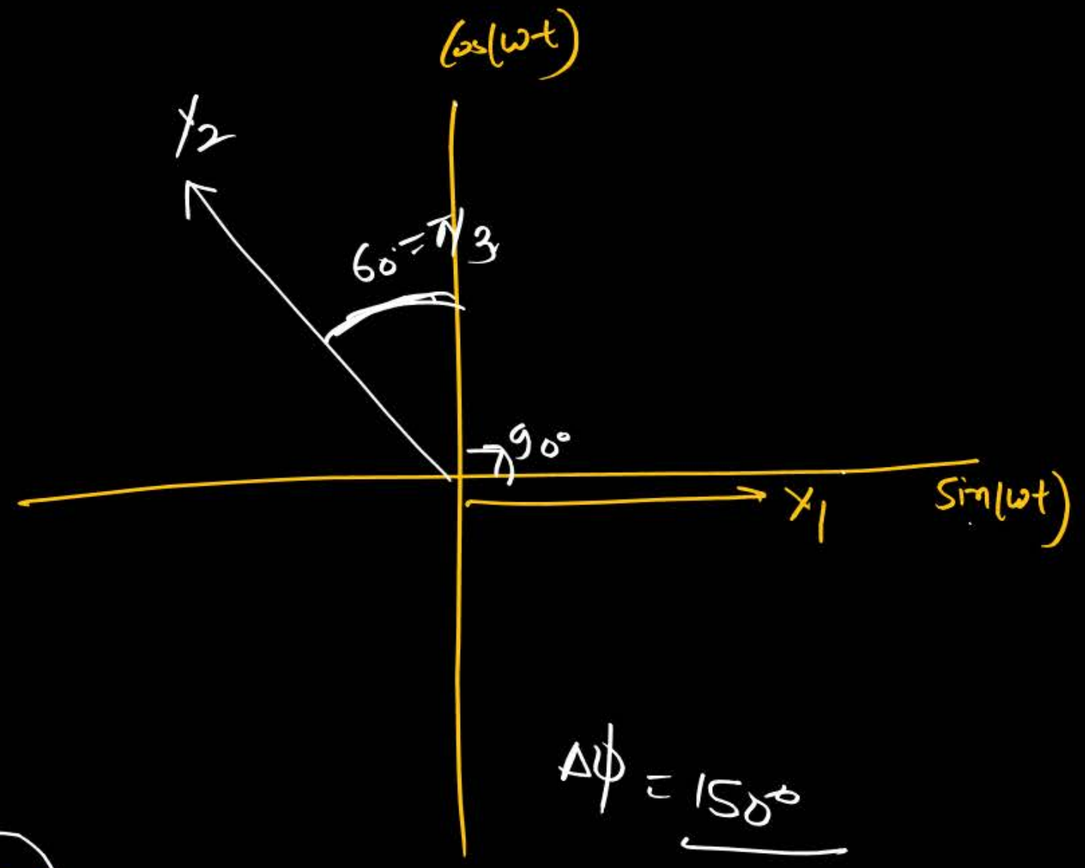
ϕ_2 आएगा

Soln

$$\Delta\phi = \phi_2 - \phi_1$$

$$= \omega t + \pi/3 - \omega t + \pi/2$$

$$= \frac{\pi}{3} + \frac{\pi}{2} = \frac{5\pi}{6} = \frac{5}{6} \times 180^{30} = 150°$$



$\Delta\phi = 150°$

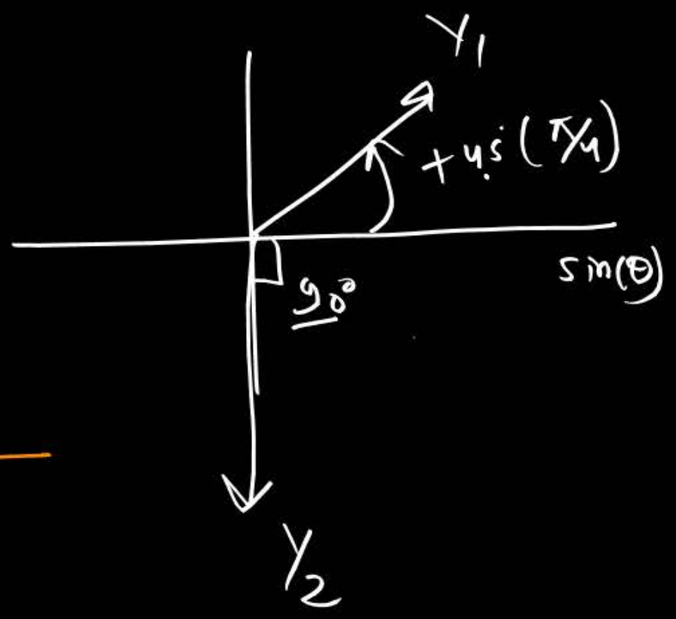
find phase difference b/w Y_1 & Y_2

① $Y_1 = 4\sin(\theta + \pi/4)$

$Y_2 = 3\sin(\theta - \pi/2)$

② $Y_1 = 4\sin(\theta + \pi/3)$

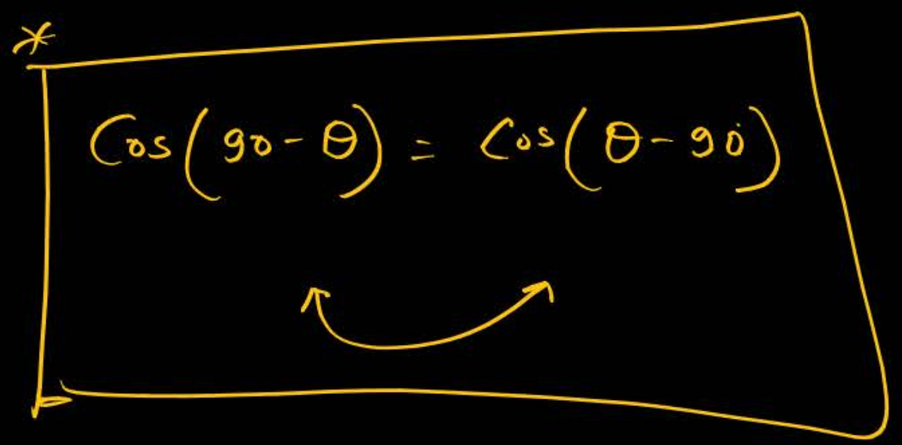
$Y_2 = 3\sin(\theta + \pi/6)$



$\Delta\phi = \phi_1 - \phi_2$

$= \theta + \pi/4 - \theta + \pi/2$

$= \frac{\pi}{4} + \frac{\pi}{3} = 135°$


$\cos(90-\theta) = \cos(\theta-90)$

find Phase difference

$$Y_1 = 3\cos(\omega t - 90^\circ) = 3\left[\cos(\omega t)\cdot\cos 90^\circ + \sin 90^\circ \sin(\omega t)\right]$$

$$= 3(0 + \sin\omega t) = 3\sin(\omega t)$$

$$Y_2 = 4\sin(\omega t)$$

$$Y_1 = 3\sin(\omega t)$$

$$Y_2 = 4\cos(\omega t + \pi/2)$$

Soln

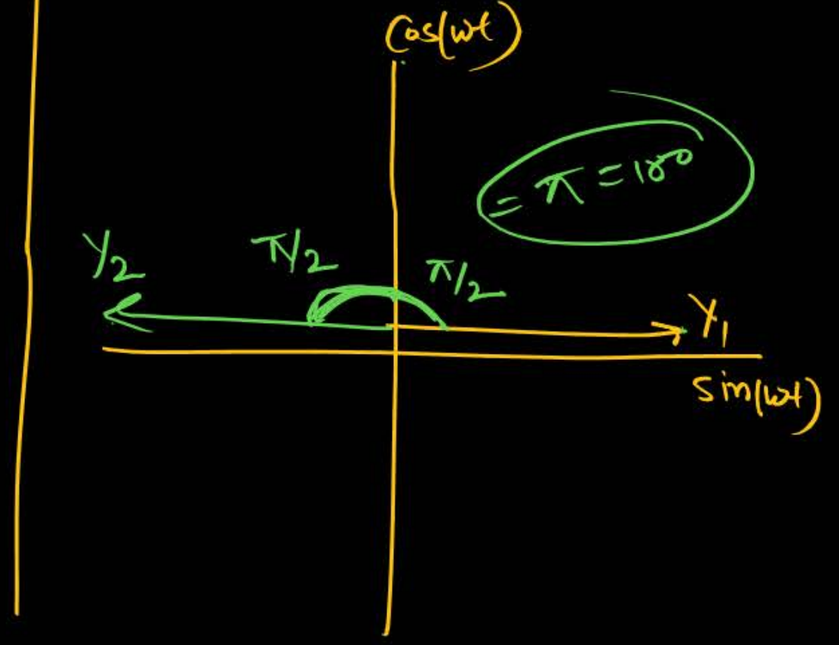
$$\Delta\phi = \phi_2 - \phi_1$$

$$= \omega t - (\omega t - 90)$$

$$= \omega t - \omega t + 90$$

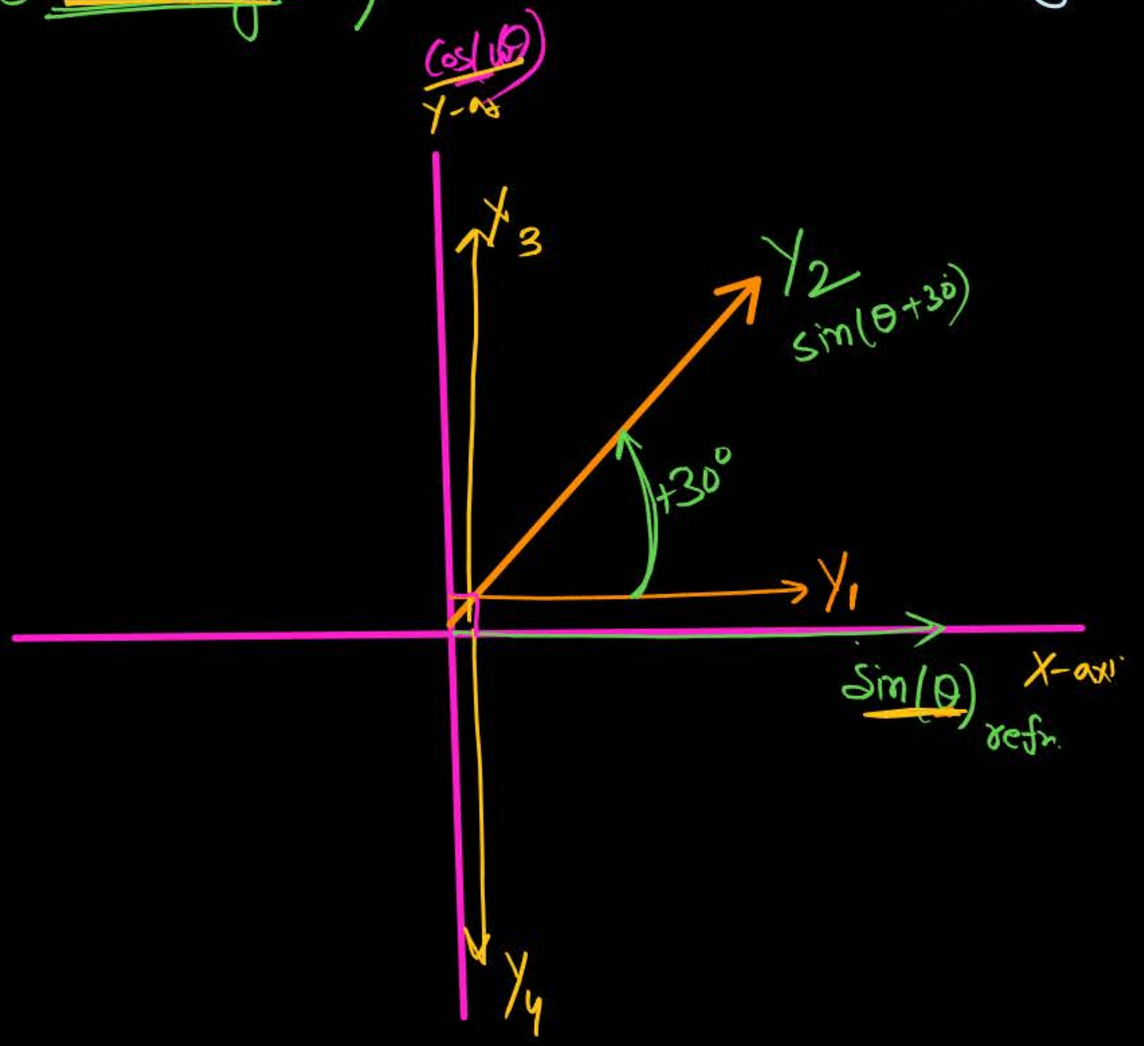
$$= 90 \quad \text{(wrong)}$$

$$\Delta\phi = 0$$



Phase diagram → Representation of Trigonometric function with Phase (Angle).

(Not a graph)



(1) $Y_1 = 4\sin(\theta)$

(2) $Y_2 = 3\sin(\theta + 30^\circ)$

↳ आगे (Anti-clock)

(3) $Y_3 = 3\sin(90+\theta) = +3\cos\theta$

(4) $Y_4 = 4\sin(\theta - 90)$

H/W

Equation – 1	Equation – 2	Phase difference
$I_1 = I_0 \sin(\theta + \pi/3)$	$I_2 = I_0 \sin(\theta - \pi/6)$	
$I_1 = I_0 \sin(\theta + \pi/3)$	$I_2 = I_0 \cos(\theta + \pi/3)$	
$I_1 = 4 \sin(\omega t)$	$I_2 = I_0\, 4 \cos(\omega t - \pi/3)$	
$I_1 = 3 \sin(\omega t + \pi/2)$	$I_2 = 5 \cos(\omega t)$	
$I_1 = 4 \cos(\omega t - \pi/2)$	$I_2 = 4 \sin(\omega t + \pi/2)$	
$I_1 = 3 \sin(\omega t + \pi/3)$	$I_2 = 3 \cos(\omega t)$	
$I_1 = 4 \sin(\theta - 60°)$	$I_2 = 4 \cos(\theta - 30°)$	

Question

Equation of a.c. current $I = -I_0 \cos(\omega t)$ and a/c voltage is $V = V_0 \sin(\omega t - \pi/6)$ then find phase difference between current and voltage:

1. $\frac{2\pi}{3}$
2. $\frac{\pi}{3}$
3. $\frac{\pi}{6}$
4. $\frac{5\pi}{6}$

Question

Current in A/C circuit is $I_1 = I_0 \sin(\omega t - 30°)$ and voltage across it $V = V_0 \cos(\omega t)$

Find phase difference?


Sanghasth Assignment

Question

What is the value of the following

(a) $\sin 30$

(b) $\sin 45°$

(c) $\sin 510°$

(d) $\sin 240°$

(e) $\sin 330°$

(f) $\cos 300°$

(g) $\tan 135°$

(h) $\cot 210°$

(i) $\cos 510°$

(j) $\sin 135°$

Question

Write the value of following:

(a) $\sin 3°$

(b) $\tan 3°$

(c) $\cos (0.5)°$

(d) $\tan 5°$

(e) $\sin 6°$

Question

What is the maximum and minimum value of the following:

(a) $4 \sin \theta + 3 \cos \theta$

(b) $2 \sin \theta - 3 \cos \theta$

(c) $1 + 5 \sin \theta$

(d) $1 - 2 \sin \theta$

(e) $5 \sin \theta - 12 \cos \theta$

Question

If $\tan\theta = \frac{5}{12}$, what is the value of $3\sin\theta + 2\cos\theta$.

Question

If $\sin\theta = \frac{\sqrt{5}}{3}$ then what is the value of the $\tan\theta + 4\sec\theta$.

Question

$\sqrt{1+\cos\theta}$ is equal to

1. $\sqrt{2}\sin\frac{\theta}{2}$
2. $\sqrt{2}\cos\frac{\theta}{2}$
3. $\frac{1}{\sqrt{2}}\sin\frac{\theta}{2}$
4. $\frac{1}{\sqrt{2}}\cos\frac{\theta}{2}$

Question

The value of $\tan^{-1}\{1\}$ is

1. $\pi/4$
2. $5\pi/4$
3. Both (1) and (2)
4. None of these

Question

A car is moving towards a building with speed 10 m/s. At any instant the angle of elevation of the building is 30° after 5 sec. the angle of elevation of the building becomes 45°, then height of building is:

1. $h = \frac{50}{\sqrt{3}+1}$
2. $h = \frac{20}{\sqrt{3}+1}$
3. $h = \frac{20}{\sqrt{3}-1}$
4. $h = \frac{50}{\sqrt{3}-1}$

Question

A person, at any instant finds angle of elevation of 40 m high building as 30°. After some time the angle of elevation becomes 60°, then value of distance, the person moved towards the building will be:

1. $d = \frac{40}{\sqrt{3}}$
2. $d = \frac{80}{\sqrt{3}}$
3. $d = \frac{60}{\sqrt{3}}$
4. $d = \frac{120}{\sqrt{3}}$

Question

What will be the displacement of a bike which moves in a circular curve of radius 30m by 2°.

1. 1.05 m
2. 60 m
3. 30 m
4. 2.1 m

Question

We can write $\cos\theta = \cos(-\theta)$ **(True/False)**

Question

$$\sin\frac{\theta}{2} = \pm\sqrt{\frac{1-\cos\theta}{2}}$$

(True/False)

Question

We can write $\tan 2\theta = \dfrac{1 - \tan^2 \theta}{2 \tan \theta}$

(True/False)

Question

We can write $\sin 2x = 2\cos^2 x - 1$ **(True/False)**

Question

$\cos(A + B) = \cos A \cos B - \sin A \sin B$ **(True/False)**

Question

Statement-1: For very small angle θ, we may use approximation $\sin\theta \approx \theta \approx \tan\theta$.
Statement-2: For very small angle θ, the hypotenuse and the base become approximately of the same length.

1. Statement-1 is true, Statement-2 is true, Statement-2 is a correct explanation for Statement-1

2. Statement-1 is true, Statement-2 is true, Statement-2 is not a correct explanation for Statement-1

3. Statement-1 is true, Statement-2 is false

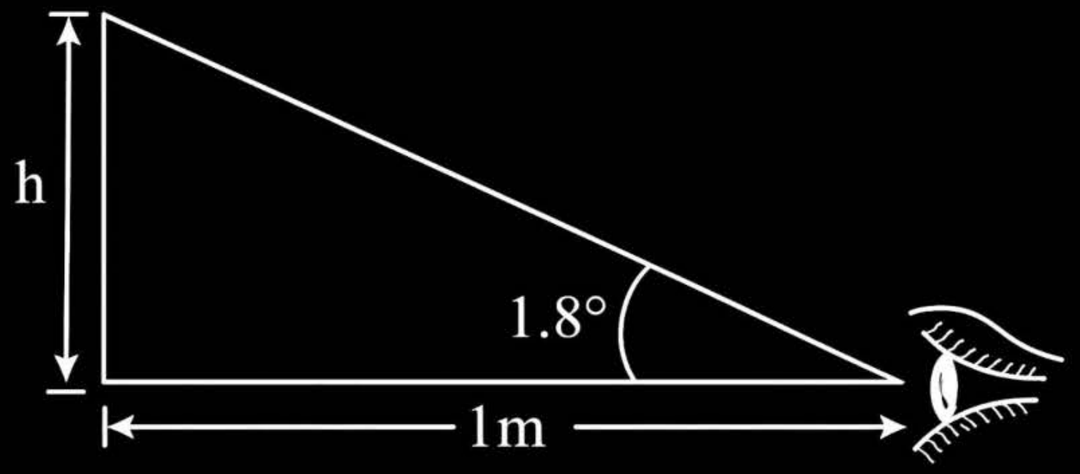
4. Statement-1 is false, Statement-2 is true

Question

What is value of expression $2(\sin 15° + \sin 75°)^2$?

Question

A normal human eye can see an object making an angle of 1.8° at the eye. What is the approximate height of object which can be seen by an eye placed at a distance of 1m from the object.



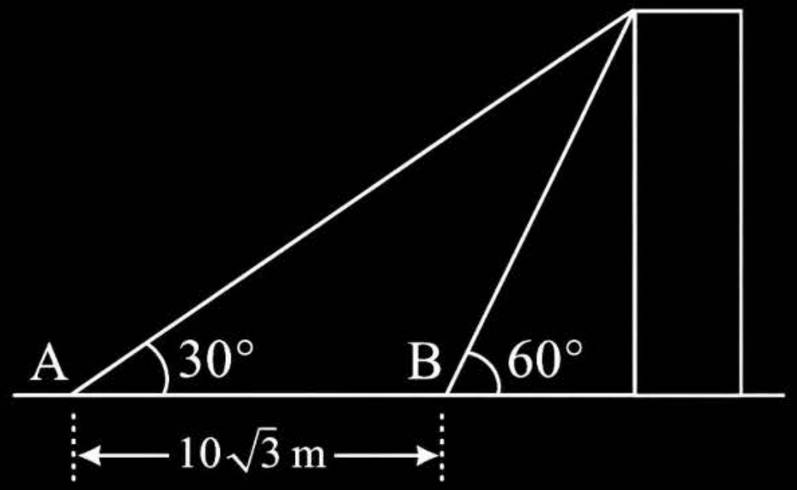
Question

The maximum and minimum values of expression $(4 - 2\cos\theta)$ respectively are:

1. 4 and 0
2. 4 and 2
3. 6 and 0
4. 6 and 2

Question

Angle of elevation is the angle which line of sight makes with the horizontal. Angle of elevation of the top of a tall building is 30° from a place A and becomes 60° from another place B that is $10\sqrt{3}$ m from. A towards the building as shown in the figure. Height of the building is close to:



1. 7.5 m
2. 10 m
3. 12.5 m
4. 15 m

Question

If $\tan\theta = \frac{24}{7}$ and $\sin\theta$ is negative then value of $\cos\theta$ will be

1. $\frac{7}{25}$
2. $-\frac{7}{25}$
3. $\frac{24}{25}$
4. $-\frac{24}{25}$

Question

Depending on in which quadrant an angle θ lies, functions $\cos\theta$ and $\sin\theta$ may be positive or negative. In the second column of the given table are specified whether these functions are positive or negative and in the first column are specified quadrants. Match Column-I with Column-II:

Column-I		Column-II	
(A)	First	**(P)**	$\sin\theta$ is positive
(B)	Second	**(Q)**	$\sin\theta$ is negative
(C)	Third	**(R)**	$\cos\theta$ and $\sin\theta$ both are negative
(D)	Fourth	**(S)**	$\sec\theta$ is negative and $\sin\theta$ is positive

Question

If $\sin\theta = 1/2$, then value of $\sqrt{3}\,(\cos\theta + 2\tan\theta)$ will be :

1. 7
2. 15/2
3. 7/2
4. None

Question

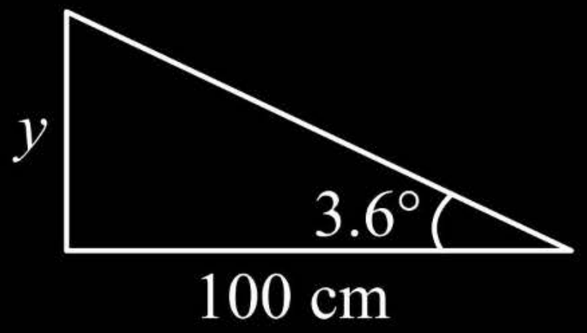
If $\tan\theta = \frac{5}{12}$, then value of $13(\cos\theta - 2\sin\theta) = ?$

1. 2
2. 1
3. 0
4. None

Question

For given triangle y = ?

1. π cm
2. $\pi/2$ cm
3. 2π cm
4. 360 cm



Question

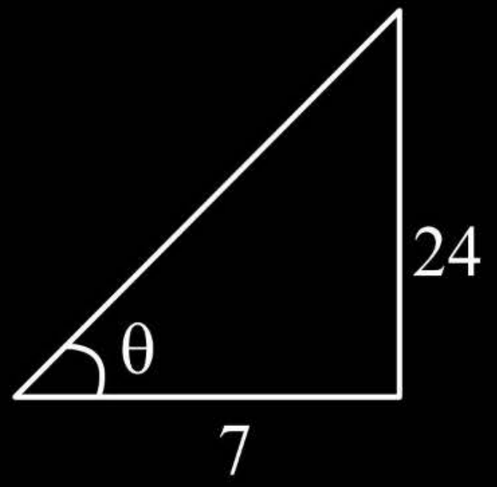
Value of $\sin\left(\frac{1.8^\circ}{\pi}\right)$ will be :

1. 0.1
2. 0.01
3. 0.001
4. 1

Question

Value of $7 \tan\theta + 100 \cos\theta - 50 \sin\theta$

1. 2
2. 4
3. 1
4. 3



Question

If $\sin\theta = 12/13$, then value of $10\tan\theta = ?$

1. 12
2. 24
3. 6
4. 12/5

Question

$\sin(-30°) + \sqrt{3}\cos(-30°) = ?$

1. 1
2. 2
3. 0
4. −1

Question

$\sin(-60°) = ?$

1. $\frac{1}{2}$
2. $-\frac{1}{2}$
3. $-\frac{\sqrt{3}}{2}$
4. $\frac{\sqrt{3}}{2}$

Question

$\cos(-45°) = ?$

1. $\frac{1}{\sqrt{2}}$
2. $-\frac{1}{\sqrt{2}}$
3. 1
4. -1

Question

$\tan(-60°) = ?$

1. $\sqrt{3}$
2. $-\sqrt{3}$
3. $\frac{1}{\sqrt{3}}$
4. $-\frac{1}{\sqrt{3}}$

Question

$\sin(150°) = ?$

1. $\frac{1}{2}$
2. $-\frac{1}{2}$
3. $-\frac{\sqrt{3}}{2}$
4. $\frac{\sqrt{3}}{2}$

Question

$2\cos^2(15°) - 1 - \cos 30° = ?$

1. 0
2. 1
3. $\sqrt{3}$
4. $\frac{1}{2}$

Question

$\cos^2 15° - \sin^2 15° = ?$

1. $\frac{3}{4}$
2. 1
3. $\frac{1}{2}$
4. $\frac{\sqrt{3}}{2}$

Question

$3\cos^2(15°) + \sin^2 15° = ?$

1. $\frac{5}{2}$
2. $\frac{4+\sqrt{3}}{2}$
3. $\frac{4-\sqrt{3}}{2}$
4. 2

Question

Value of angle $\frac{\pi}{6}$ rad, $\frac{\pi}{4}$ rad, $\frac{2\pi}{3}$ rad and $\frac{5\pi}{6}$ rad in degree will be respectively :

1. 30°, 45°, 150°, 12°
2. 30°, 45°, 120°, 150°
3. 60°, 45°, 150°, 120°
4. 60°, 45°, 120°, 150°

Question

Value of angle 60°, 90°, 180°, 135° in radian will be respectively:

1. $\frac{\pi}{3}, \frac{\pi}{4}, \pi, \frac{3\pi}{4}$
2. $\frac{\pi}{4}, \frac{\pi}{3}, \pi, \frac{3\pi}{4}$
3. $\frac{\pi}{3}, \frac{\pi}{2}, \pi, \frac{3\pi}{4}$
4. $\frac{\pi}{6}, \frac{\pi}{2}, \pi, \frac{3\pi}{4}$

Do H/W in noter

→ Sanghavi Assignment in rough.

Thank

You